Long-range epigenetic concordance

a per-cell, per-bin signal that separates cell states even when mean methylation / accessibility is identical

Single-cell NOMe-HiC contact

chromatin contact

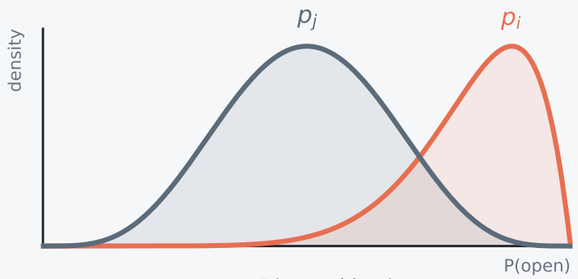


locus *i* — locus *j*

k_i/n_i — k_j/n_j

GCH accessibility · HCG (CpG) methylation

Per-end state posterior



$$p = P(\text{open} \mid k, n)$$

Beta–Binomial mixture (EM)

Soft pairwise co-state



co-open

$p_i\, p_j$

co-closed

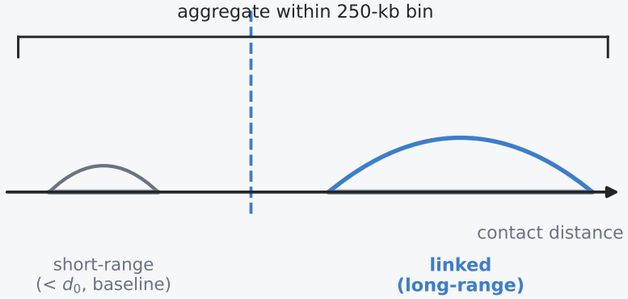
$(1-p_i)(1-p_j)$

mixed

$p_i(1-p_j) + (1-p_i)p_j$

soft expected counts (no thresholding)

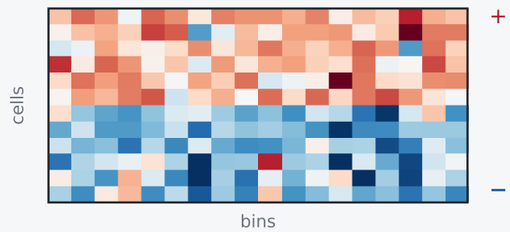
Long-range vs local baseline



Signed long-range concordance

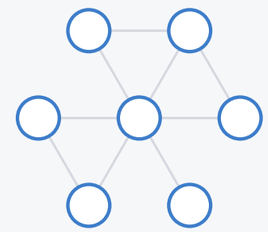
$$\Delta_{\text{signed}} = (\hat{\pi}_{\text{co-open}} - \hat{\pi}_{\text{co-closed}})_{\text{linked}}$$

$$-\,(\hat{\pi}_{\text{co-open}} - \hat{\pi}_{\text{co-closed}})_{\text{short-range}}$$



Cell-graph denoising (RWR)

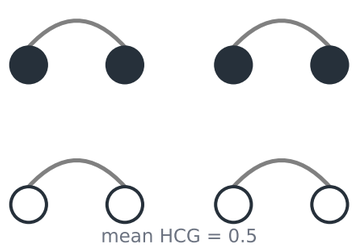
random-walk-with-restart smoothing
over the cell *k*-NN graph



quantile-normalised, imputed → UMAP

Result: identical mean methylation / accessibility, opposite long-range concordance

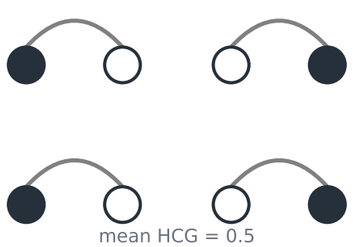
cell state 1



mean HCG = 0.5

concordance = 1.0

cell state 2

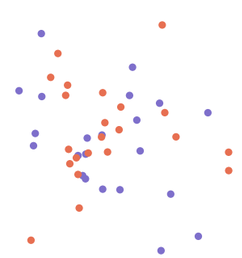


mean HCG = 0.5

concordance = 0.0

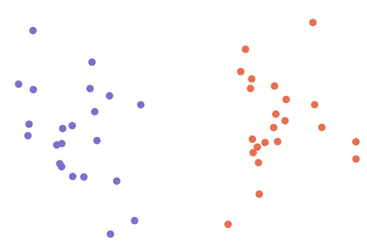
workflow

UMAP: mean HCG / GCH



states overlap — not separable

UMAP: signed concordance



states separate

methylated / accessible — unmethylated / closed

each loop = two loci in contact within one cell